

PAPER_790: ESO 391-12 — Three-UQFF Lenticular Galaxy with Dust Ring

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Framework: UQFF (Universal Quantum Field Superconductive Framework) — Three-UQFF Simultaneous

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Abstract

ESO 391-12 is a lenticular galaxy ~ 100 million light-years distant ($z \approx 0.0067$) in the constellation Centaurus, notable for a well-defined outer dust ring structure visible in Hubble imaging. Like NGC 5866 (PAPER_783) and NGC 7049 (PAPER_779), ESO 391-12 belongs to the class of quiescent early-type galaxies with preserved dust features from past gas-rich events. Its larger effective radius ($r \approx 50$ kly $= 4.73 \times 10^{20}$ m) at the same distance as NGC 7049 suggests a more extended, less concentrated stellar body. Three-UQFF analysis yields $g_{\text{primary}} \approx 1.053 \times 10^{-3} \text{ m/s}^2$ across all three modes, consistent with the quiescent lenticular class.

1. Introduction

ESO 391-12's dust ring extends well beyond its central stellar body, indicating an external gas accretion event (minor merger or tidal interaction) that delivered metal-rich gas to the outskirts. This ring is relatively tenuous and shows only very low SFR ($\sim 0.1 \text{ M}_{\odot}/\text{yr}$). Its extended distribution ($r > \text{NGC 7049}$ at similar distance) indicates a lower central stellar concentration. Three-UQFF probes whether the extended radius changes the mode convergence result — confirming that g_{primary} remains robust at the standard quiescent lenticular value.

2. Three-UQFF Parameters

Parameter	Symbol	Value	Source
Galaxy mass	M	$10^{11} \text{ M}_{\odot} = 1.989 \times 10^{41} \text{ kg}$	Estimate
Effective radius	r	$4.73 \times 10^{20} \text{ m}$ (~ 50 kly)	HST
SFR	—	$0.1 \text{ M}_{\odot}/\text{yr}$	Dust ring only
Age	t	$5 \times 10^9 \text{ yr} = 1.578 \times 10^{17} \text{ s}$	Hubble time
M_sf	—	0.008	UQFF minimal
Redshift	z	0.0067	Spectroscopic
v_EM	v	10^5 m/s	Disk rotation
B_EM	B	10^{-5} T	Quiescent field

3. Three-UQFF Derivation

Mode 1: Compressed UQFF

$$\begin{aligned} g_{\text{grav}} &= 6.6743\text{e-}11 \times 1.989\text{e}41 / (4.73\text{e}20)^2 \\ &= 1.328\text{e}31 / 2.237\text{e}41 = 5.936\text{e-}11 \text{ m/s}^2 \\ H(z) \times t &= 2.34\text{e-}18 \times 1.578\text{e}17 = 0.369; \text{ factor} = 1.369 \\ \text{factor_sf} &= 1.008; \text{ factor_TRZ} = 1.02 \\ g_{\text{grav_total}} &= 5.936\text{e-}11 \times 1.369 \times 1.008 \times 1.02 = 8.360\text{e-}11 \text{ m/s}^2 \\ a_{\text{EM}} &= 1.053\text{e-}3 \text{ m/s}^2 \\ g_{\text{comp}} &= 1.053\text{e-}3 \text{ m/s}^2 \end{aligned}$$

Mode 2: Resonant UQFF

$$g_{\text{res}} = 1.053\text{e-}3 \times 1.000285 = 1.053\text{e-}3 \text{ m/s}^2$$

Mode 3: Buoyancy UQFF

$$\begin{aligned} V &= (4/3)\pi(4.73\text{e}20)^3 = 4.44\text{e}62 \text{ m}^3; a_{\text{Ubi}} \ll a_{\text{EM}} \\ g_{\text{buoy}} &= 1.053\text{e-}3 \text{ m/s}^2 \end{aligned}$$

Three-UQFF Simultaneous Result

$$\begin{aligned} g_{\text{compressed}} &= 1.053\text{e-}3 \text{ m/s}^2 \\ g_{\text{resonant}} &= 1.053\text{e-}3 \text{ m/s}^2 \\ g_{\text{buoyancy}} &= 1.053\text{e-}3 \text{ m/s}^2 \\ g_{\text{primary}} &= 1.053\text{e-}3 \text{ m/s}^2 \end{aligned}$$

4. Physical Interpretation

ESO 391-12 at $r = 4.73 \times 10^{20}$ m has a slightly smaller g_{grav} ($5.936 \times 10^{-11} \text{ m/s}^2$) than NGC 7049 at $r = 5 \times 10^{20}$ m ($5.311 \times 10^{-11} \text{ m/s}^2$) due to the inverse-square law offsetting somewhat different radii and mass. Both remain far below the UQFF EM term. The dust ring, while photometrically prominent in HST imaging, contributes negligibly to the mass budget (dust is $\sim 1\%$ of the total interstellar medium, which itself is $\sim 1\%$ of stellar mass). Three-UQFF mode convergence is confirmed regardless of the dust ring extent.

5. Conclusions

Three-UQFF applied to ESO 391-12 yields $g_{\text{primary}} \approx 1.053 \times 10^{-3} \text{ m/s}^2$ across all three modes. Extended dust ring at ~ 50 kly radius has no net effect on UQFF mode convergence; quiescent lenticular result is confirmed.

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